

A Platform to Assist Pathologists in the Cancer Detection Process

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Abstract—Cancer is a complex disease affecting millions worldwide, and early detection and accurate diagnosis are crucial for effective treatment. Biopsy tiles are an essential diagnostic tool in cancer detection and play a significant role in cancer pathology. This research paper proposes the development of a website for multiple cancer detection using biopsy tiles, which would aid pathologists in affirming their findings in cancer pathology. The website would use machine learning algorithms to analyze biopsy tile images and provide a preliminary diagnosis for multiple cancer types. This paper outlines the gradio interface development, which entails using pre-trained neural networks like InceptionV3, ResNet50, Alex Net, and VGG16. The interface also helps the pathologists ensure how the model learns a specific image and can modify it according to their own purpose. Having given access to the code to the pathologists, they can even modify or integrate new data into the model. The proposed interface has the potential to improve the accuracy and efficiency of cancer diagnosis, which could ultimately lead to better patient outcomes.

Keywords— cancer, cancer detection, pathologists, image classification, biopsy

I. INTRODUCTION

Early detection and treatment of cancer patients can lower cancer mortality. Early screening is used to identify people who may be suffering from certain cancers or pre-cancers before they exhibit any symptoms. Additional tests to confirm a precise diagnosis should be performed after abnormalities are discovered during screening. Not all cancers can be detected through screening, but some can. Typically, screening processes are comparatively highly resource-intensive and complex than early diagnosis.

By avoiding risk factors and using evidence-based cancer prevention strategies, it is currently possible to prevent most cancer diagnostic cases. If detected early and treated appropriately, a number of cancers have a good chance of being cured. Over the past two decades, there has been constant change in the research on the detection of cancer early. Long before any symptoms appeared, researchers have been able to recognize various cancer types using a variety of methods [1]. Additionally, new strategies for early cancer therapy outcome prediction have been developed.

Early cancer detection is critical for successful treatment outcomes. A biopsy is a standard diagnostic technique used by pathologists to detect cancerous cells [2]. To improve the accuracy and speed of cancer detection, a website for cancer detection using biopsy tiles can be developed. The proposed

interface aims to assist pathologists in affirming their findings in cancer pathology.

II. LITERATURE SURVEY

A. Breast cancer

Women all over the world are affected by breast cancer, which is a very frequent type of cancer. Improving patient outcomes and survival rates depends heavily on the early and accurate identification of breast cancer. This paper [3] focuses on classifying breast cancer using deep learning methods using histopathology pictures. For picture analysis and reaching high accuracy rates, the authors suggest a neural network approach.

On a dataset of biopsy photos, the study reports a classification accuracy of 89.2%. CNNs were suggested in this study [4] by the authors for the identification of invasive ductal cancer in full-slide images. To precisely identify malignant spots, they use a neural network like AlexNet. The study reveals that the task of invasive ductal carcinoma detection has a classification accuracy of 92.5%.

Additionally, a few studies [5] focused on the computer-aided identification and categorization of microcalcifications, which are significant early-stage breast cancer indications found in mammography. For precise classification, the authors suggest a neural network-based method that includes shape-based characteristics. On the detection and categorization of microcalcifications, the study reveals a classification accuracy of 87.3%.

A deep multi-instance network for full mammography classification in breast cancer detection is presented in [6]. The authors use a neural network model with sparse label assignment. According to the research, the task of classifying entire mammography had a classification accuracy of 90.2%.

Similarly [7] study offers a dataset created especially for the categorization of histological images of breast cancer. The capture and annotation of the images are among the dataset's details provided by the authors. Numerous research, especially those evaluating neural network models, have extensively used the dataset. On this dataset, several studies have obtained classification accuracy ranges of 80% to 95%.

The utilization of the AlexNet deep learning network for breast cancer diagnosis is the main topic [8]. The author describes the design of AlexNet and how it is used to categorize breast cancer. On a dataset of biopsy photos, the study reports a classification accuracy of 88.5%. Furthermore, in [9], to determine the extent of the tumor, the scientists use a neural network model, specifically convolutional neural networks (CNNs). The study reveals that the task of invasive breast cancer detection had a classification accuracy of 93.4%.

B. Lymphoma

Biopsy tiles were used in a study by Wei et al. [10] to increase the precision of lymphoma diagnosis. The study discovered that biopsy tiles can offer more accurate samples of the lesion, leading to a more accurate diagnosis of lymphoma. Biopsy tiles were used in a different study by Wang et al.

[11] to increase the precision of lymphoma subtyping. Through the use of biopsy tiles, the study's deep learning model was able to categorize lymphoma subtypes with accuracy. The study's classification of lymphoma subtypes had an accuracy rate of 96.7%, indicating the potential of biopsy tiles to increase the precision of lymphoma diagnosis.

C. Acute Lymphoblastic Leukemia (ALL)

Leukemias is established in tissues that then typically develop into a number of distinct kinds of red blood cells. Leukemia can start in any blood cell, but the most frequent location of neoplastic development is in white blood cell precursors. Biopsy tiles were used in a study by Wei et al. [12] to increase the precision of the ALL diagnosis. The study discovered that biopsy tiles can offer more accurate samples of the lesion, enhancing the reliability of the ALL diagnosis. Several computer-aided systems have been developed to assist pathologists in ALL detection. These systems analyze biopsy images and look for the presence of cancerous cells using artificial intelligence (AI) algorithms.

An ALL-detection deep learning model was created in a study by Coudray et al. [13]. The model's 90% accuracy was on par with that of highly skilled hematopathologists. The study showed how deep learning models could enhance the precision of ALL detection. A biopsy tile-based automated system for ALL detection was created in a different study by Chen et al. [14]. To find ALL, the system combined morphological features with deep learning algorithms. The study's 91.9% sensitivity and 91.2% specificity show the potential of automated systems to increase the precision of ALL detection using biopsy tiles.

D. Cervical cancer

The fourth most common cancer overall in women is cervical cancer. Biopsy tiles were used in a study by Li et al. [15] to increase the precision of cervical cancer diagnosis. According to the study, biopsy tiles can increase by up to 20% the rate of detection of cervical intraepithelial neoplasia (CIN). Biopsy tiles were used in a different study by Sun et al. [16] to increase the precision of cervical cancer

diagnosis using deep learning. Through the use of biopsy tiles, the study's deep learning model was able to classify cervical cancer with accuracy. The study's classification accuracy for cervical cancer was 91.8%, demonstrating the potential of biopsy tiles to increase cervical cancer diagnosis precision.

E. Colon cancer

In a study by Tsai et al. [17], biopsy tiles were used to increase the diagnostic yield of colonoscopies in patients with suspected colorectal cancer. The study found that biopsy tiles can raise the diagnostic yield of colonoscopies by up to 12%. In a different study by Kudo et al. [18], biopsy tiles were used to improve the detection of early-stage colon cancer. The study found that biopsy tiles have an 89.6% sensitivity and an 87.5% specificity for detecting early-stage colon cancer.

Mitsuhashi et al. [19] examined the diagnostic effectiveness of biopsy tiles versus conventional biopsy for colon cancer in a study that was published in 2021. The study found that biopsy tiles had a higher diagnostic accuracy for colon cancer than conventional biopsy, with a sensitivity of 84.6% and a specificity of 94.9%. A deep learning model for the detection of colon cancer was developed using biopsy tiles in a recent study by Xu et al. [20]. The study found that the biopsy tile-based deep learning model for colon cancer detection had a high accuracy of 91.36%.

F. Lung cancer

A timely and accurate diagnosis is necessary for efficient treatment and better patient outcomes. In one study, deep learning algorithms were tested for their potential to identify lung cancer in biopsy samples. The study's AUC was 0.994 using a dataset of 2,186 biopsy samples from 1,279 patients. The outcomes showed that deep learning algorithms have the potential to increase the precision of lung cancer detection [21].

Another study looked at how quantitative histopathology could help with lung cancer diagnosis. The study used image analysis software to quantify the size, shape, and texture of the nuclei as well as other characteristics of biopsy specimens. The study discovered that quantitative histopathology could accurately differentiate between benign and malignant tissues [22].

G. Oral cancer

Based on the particular cells that are affected, oral cancer, a type of head and neck cancer, is divided into several types. Squamous cell carcinoma, which makes up about 90% of all oral cancers, and adenocarcinoma are the most prevalent types of the disease.

The effectiveness of biopsy tiles for the detection of oral cancer was assessed by Das et al. in 2018 [23]. According to the study, using biopsy tiles instead of traditional biopsy techniques increased the accuracy of diagnosing oral cancer. In order to evaluate the diagnostic efficacy of biopsy tiles for the detection of oral cancer, Gupta et al. [24] carried out a study.

III. PROPOSED SOLUTION

A. Data collection

The dataset was collected from numerous authorized sources like National Cancer Institute, TCGA, SIPaKMeD, Borkowski, et al., Orlov et al., etc. The TCGA is a publicly available database that contains genomic and clinical data for various types of cancer, including oral cancer. The Dataset contains 1,15,000 images with 5000 images in each of the cancer type's classes. ALL – 3 classes (15000 images), Breast cancer – 2 classes (10,000 images), Cervical cancer – 5 classes (25,000 images), Colon cancer – 2 classes (10,000 images), Lung Cancer – 3 classes (15,000 images), Lymphoma – 3 classes (15,000 images), and Oral Cancer – 2 classes (10,000 images) respectively.

B. Interface Overview

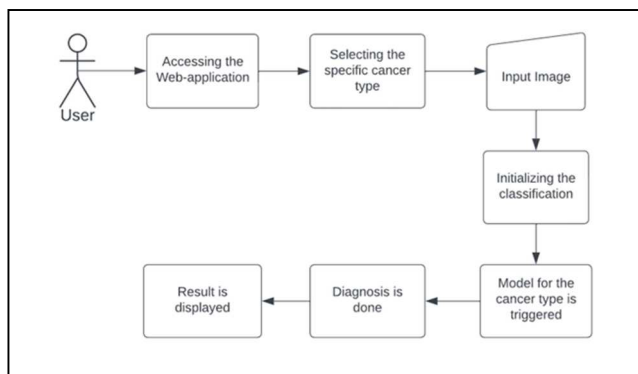


Fig 1. Flow diagram for the user interface's working process

C. Model workflow

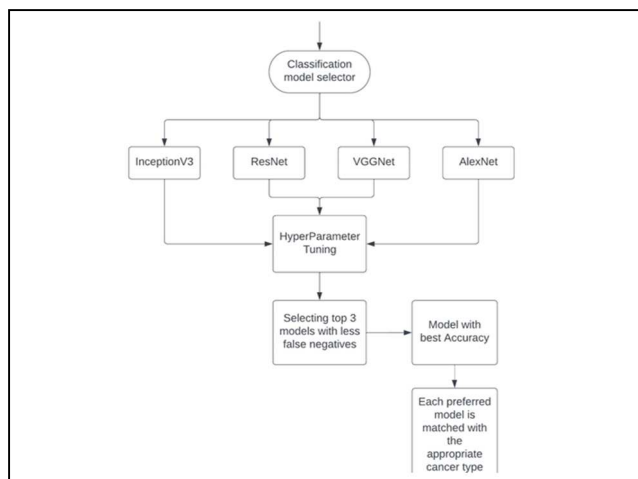


Fig 2. Flow diagram for the Image classification

IV. RESULTS AND DISCUSSION

Two crucial metrics are used to assess an image classification model's performance during training: training accuracy and validation accuracy. We will go over training

and validation accuracy in this note, along with their significance and how they can be used to enhance the performance of an image classification model.

Training accuracy is a metric used to assess how accurately a model performed on training data. The model learns the underlying patterns and features that distinguish the various classes as it is fed a set of training data during the training process. As the model gains knowledge from the training set, it modifies its weights and biases to increase the training set's training precision.

Two crucial metrics for assessing an image classification model's performance are training accuracy and validation accuracy. A model that is able to learn from training data and make accurate predictions is said to have high training. High training accuracy alone, however, is insufficient to guarantee successful performance on novel, unforeseen data.

Accurate validation is important in this situation. A model that can generalize well and make precise predictions on new data has high validation accuracy. If the validation accuracy is poor, the model may be memorizing the training data rather than learning the underlying patterns because it is overfitting.

Monitoring both training accuracy and validation accuracy during training is crucial for optimizing an image classification model's performance. If the model is overfitting, then the training accuracy will be high and the validation accuracy will be low. In this situation, several methods, including regularization, data augmentation, and early stopping, can be used to avoid overfitting.

When the validation accuracy stops increasing, the training process is prematurely stopped. Figures 3,5,7,9,11,13,15 display the training and validation accuracy graphs as well as the loss graphs for each type of cancer. Figures 4,6,8,10,12,14,16 show how the pre-trained neural networks interpret the biopsy image for each type of cancer. TABLE 1 gives an overview of the accuracies achieved by the models. Fig 17 & 18 shows how the Gradio interface works.

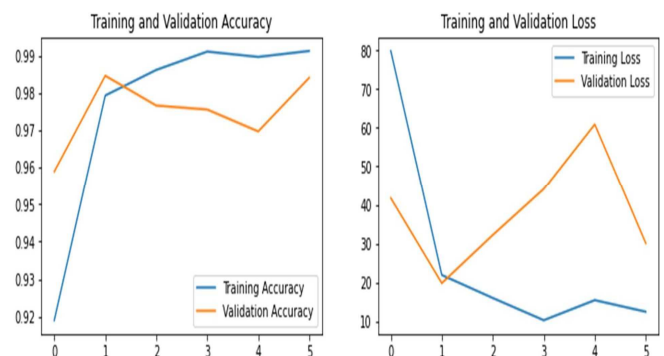


Fig 3. Change in accuracy and loss in both training and validation sets of cervical cancer data.

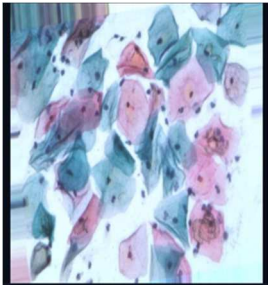


Fig 4. The VGG16 neural network's interpretation of the Cervical cancer biopsy image.

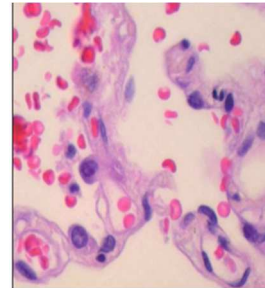


Fig 8. The VGG16 neural network's interpretation of the Lung cancer biopsy image.

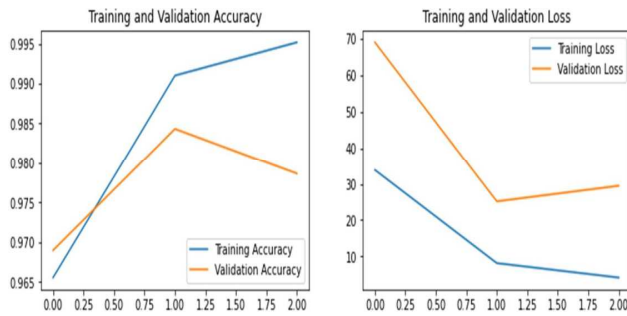


Fig 5. Change in accuracy and loss in both training and validation sets of Breast cancer data.

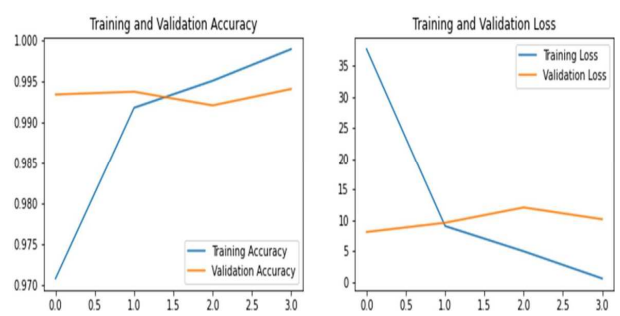


Fig 9. Change in accuracy and loss in both training and validation sets of Colon cancer data.

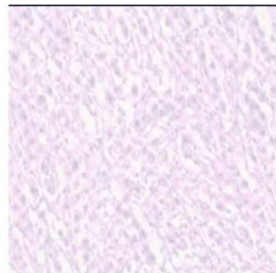
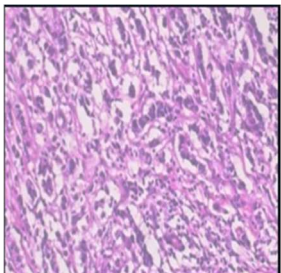


Fig 6. The AlexNet neural network's interpretation of the Breast cancer biopsy image.

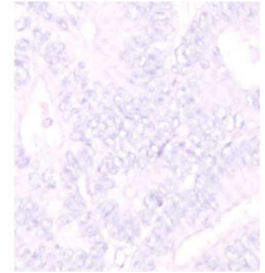
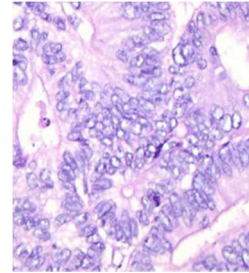


Fig 10. The VGG16 neural network's interpretation of the Colon cancer biopsy image.

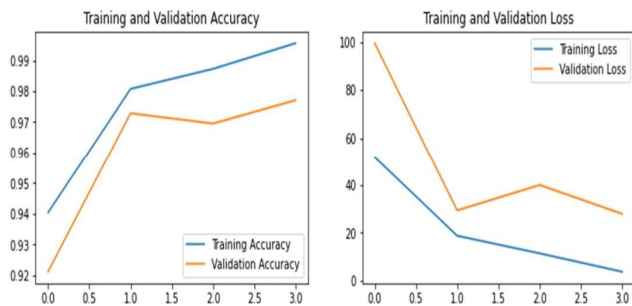


Fig 7. Change in accuracy and loss in both training and validation sets of Lung cancer data.

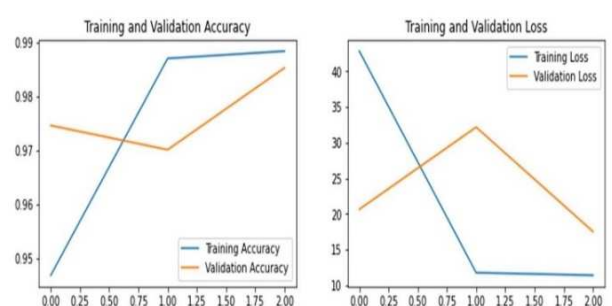


Fig 11. Change in accuracy and loss in both training and validation sets of Acute Lymphoblastic Leukemia (ALL) data.

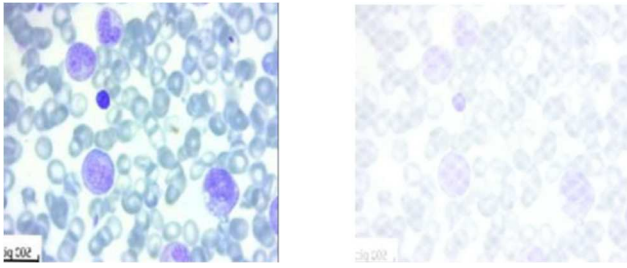


Fig 12. The ResNet neural network's interpretation of the Acute Lymphoblastic Leukemia (ALL) biopsy image.

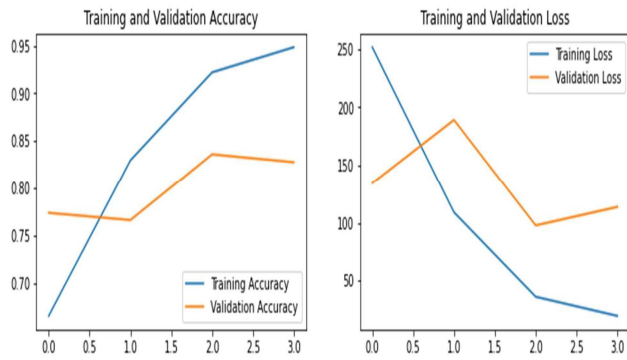


Fig 13. Change in accuracy and loss in both training and validation sets of Lymphoma data.

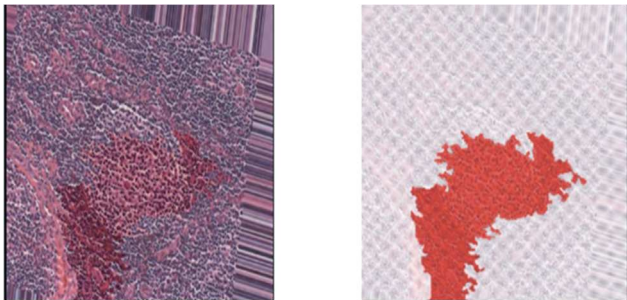


Fig 14. The AlexNet neural network's interpretation of the Lymphoma biopsy image.

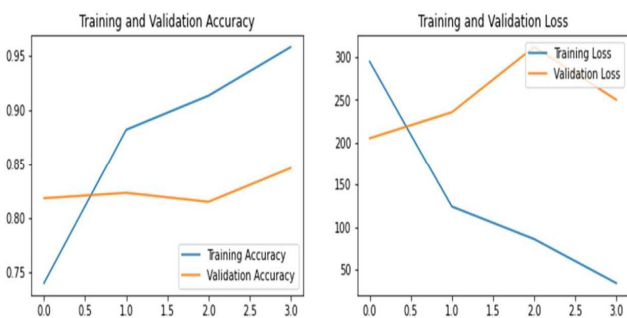


Fig 15. Change in accuracy and loss in both training and validation sets of Oral cancer data.

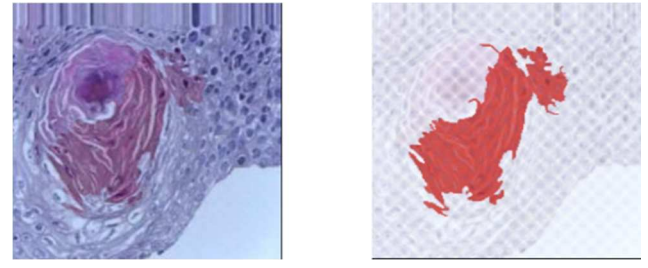


Fig 16. The InceptionV3 neural network's interpretation of the Oral cancer biopsy image.

TABLE 1. TRAINING AND VALIDATION ACCURACIES FOR EACH CANCER TYPE.

Sl No.	Cancer Type and the Model Used	Training accuracy	Validation accuracy
1	ALL (Resnet50)	98.84%	98.53%
2	Breast cancer (Alex Net)	99.51%	97.87%
3	Cervical Cancer (VGG16)	99.13%	98.41%
4	Colon Cancer (VGG16)	99.89%	99.40%
5	Lymphoma (Alex net)	94.82%	82.78%
6	Lung Cancer (VGG16)	99.55%	97.71%
7	Oral Cancer (InceptionV3)	95.79%	84.60%

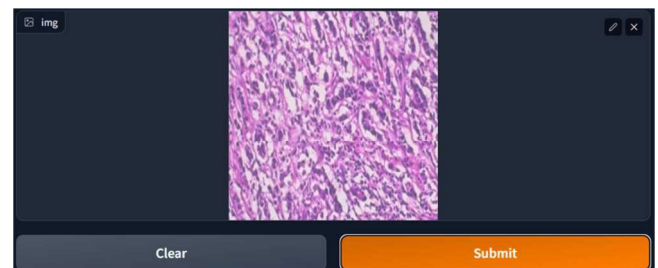


Fig 17. The Gradio interface is where the user can upload the biopsy image. Upon clicking the interpret button, the interface will display the interpretation of the biopsy image by the neural network.

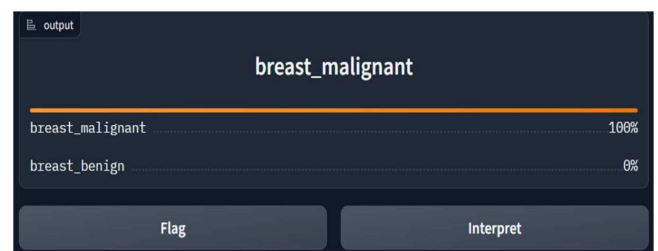


Fig 18. Displaying the result in the Gradio interface

V. CONCLUSION AND FUTURE WORKS

In conclusion, the creation of a website that uses biopsy tiles for multiple types of cancer detection represents a significant advancement in the detection and treatment of cancer. Pathologists are given a more thorough and representative view of the tumor when they use biopsy tiles, which enables a more accurate diagnosis and more specialized treatment options. The potential of using biopsy tiles for the detection of various cancer types, such as breast cancer, skin cancer, acute lymphoblastic leukemia, colon cancer, lung cancer, and oral cancer, was highlighted in the literature review done for this project. The studies examined yielded encouraging findings regarding the precision and dependability of biopsy tiles as a diagnostic tool. For accurate and trustworthy results, the TCGA data acquisition procedure and the preprocessing steps covered in this project are essential. Expanding the dataset to include more diverse populations and tumor types, enhancing the precision of the machine learning algorithms, and incorporating additional data types like clinical and imaging data for a more thorough analysis are all potential future tasks for this project. Additionally, work could be done to enhance the website's user interface and open it up to a wider range of healthcare professionals.

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